

Gnana Deepak Madduri

Data Engineer

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EDUCATION

Masters from Western Michigan University, USA

TECHNICAL SKILLS

- **Hadoop/Spark Ecosystem:** Hadoop, MapReduce, Pig, Hive/impala, YARN, Kafka, Flume, Sqoop, Oozie, Spark, Airflow, MongoDB, Cassandra, HBase, and Storm.
- **Programming Languages:** Java, Python, Hibernate, JDBC, JSON, HTML, CSS.
- **Cloud Technologies:** AWS, GCP, Amazon, S3, EMR, Redshift, Lambda, Athena Composer, Big Query.
- **Script Languages:** Python, Shell Script (bash, shell).
- **Databases:** Oracle, MySQL, SQL Server, PostgreSQL, HBase, Snowflake, Cassandra, MongoDB.
- **Version controls and Tools:** GIT, Maven, SBT, CBT.
- **Web/Application server:** Apache Tomcat, WebLogic, WebSphere
- **AWS Ecosystem:** S3Bucket, Athena, Glue, EMR, Redshift, Data Lake, AWS Lambda, Kinesis
- **Azure Ecosystem:** Azure DataLake, ADF, Databricks, Azure SQL
- **Operating Systems:** Windows, Unix, Linux

CAREER OBJECTIVE

Highly qualified Data Engineer with **5+ years** of experience in the industry. Enjoy creative problem solving and getting exposure on multiple projects, and I would excel in the collaborative environment on which your company prides itself.

PROFESSIONAL SUMMARY

- Around 5+ years of experience in **Data engineering** encompassing Requirements **Analysis, Design Specification, and Testing** in both **Waterfall** and **Agile** methodologies.
- Hands-on experience with **Amazon EC2, Amazon S3, Amazon RDS, VPC, IAM, Amazon Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SES, SQS, Lambda, EMR** and other services of the **AWS** family.
- Worked with **csv, Avro, Parquet** data to load into **Data frames** and do the analysis. Hands on experience on **Google Cloud Platform (GCP)** in all the bigdata products **BigQuery, Cloud Data Proc, Google Cloud Storage, Composer (Air Flow as a service)**.
- Proficiency with **Scala, Apache HBase, Hive, Pig, Sqoop, ZooKeeper, Spark, Spark SQL, Spark Streaming, Kinesis, Airflow, Yarn, and Hadoop (HDFS, MapReduce)**. Implemented **Big Data solutions** using **Hadoop** technology stack, including **Pyspark, Hive, Sqoop, Avro** and **Thrift**.
- Working knowledge of **HDFS, Kafka, Map Reduce, Spark, PIG, HIVE, Sqoop, HBase, Flume, and Apache ZooKeeper** as tools for designing and deploying end-to-end big data ecosystems.
- Hands-on experience with **Spark, Databricks, and Delta Lake**.
- Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Experience in automating day-to-day activities by using **Windows PowerShell**.
- Experienced working in both **Agile** and **Waterfall** based development environment and participating in **Scrum** sessions.
- Experience in implementing **Azure** data solutions, provisioning storage account, **Azure Data Factory, SQL server, SQL Databases, SQL Data warehouse, Azure Data Bricks** and **Azure Cosmos DB**.
- Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Experienced in building **Snow Pipes**, migrating **Teradata** objects into **Snowflake** environment.
- Proficient with Container systems like **Docker** and Container orchestration like **EC2 Container Service, Kubernetes**, worked with **Terraform**.
- Expert in designing Parallel jobs using various stages like **Join, Merge, Lookup, remove duplicates, Filter, Dataset, Lookup file set, Complex flat file, Modify, Aggregator, XML**.
- Hands-on experience using **Snowflake** utilities, **Snow SQL, Snow Pipe, Big Data model techniques using Python / Java**.
- Expertise in building **CI/CD** on **AWS** environment using **AWS Code Commit, Code Build, Code Deploy** and **Code Pipeline** and experience in using **AWS CloudFormation, API Gateway**, and **AWS Lambda** in automation and securing the infrastructure on **AWS**.

WORK EXPERIENCE

Client: Ally Financial, Detroit, Michigan, USA

Sep 2023 - Present

Role: Azure Data Engineer

Description: Ally Financial is a bank holding company. It provides financial services. Developed and optimized data systems for ingestion, transformation, and real-time analytics, enhancing overall data processing and business intelligence capabilities.

Responsibilities:

- Designed and develop **JAVA API** (Commerce **API**) which provides functionality to connect to the **Cassandra** through **Java** services. Designed **SSIS** control flow tasks for orchestrating the sequence and logic of **ETL** processes, such as conditional branching, looping, and error handling.
- Involved in Configuring and Upgrading the On Premises Data gateway between various Data sources like **SQL Server** to **Azure Analysis Services** and **Power BI** service. Used **Spark Streaming** to receive real time data from the **Kafka** and store the stream data to **HDFS** using **Python** and **NoSQL** databases such as **HBase** and **Cassandra**
- Load and transform large sets of structured, semi structured, and unstructured data using **Hadoop/Big Data concepts**.
- Wrote **Spark-Streaming** applications to consume the data from **Kafka** topics and write the processed streams to **HBase**.
- Experience in Developing **Spark** applications using **Spark - SQL** in **Databricks** for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns. Responsible for estimating the cluster size, monitoring, and troubleshooting of the **Spark Databricks cluster** and Ability to apply the spark **Data Frame API** to complete **Data** manipulation within **spark session**.
- Created **Data tables** utilizing **PyQt** to display customer and policy information and add, delete, update customer records.
- Developed custom **UDFs** in python to be used in **spark-SQL**. Utilized **Elasticsearch** and **Kibana** for indexing and visualizing the real-time analytics results, enabling stakeholders to gain actionable insights quickly.
- Involved in various phases of Software Development Lifecycle (**SDLC**) of the application, like gathering requirements, design, development, deployment, and analysis of the application.
- Wrote data ingestion systems to pull data from traditional **RDBMS** platforms such as **Oracle** and **Teradata** and store it in **NoSQL** databases such as **MongoDB**.
- Developing and maintaining **Azure Analysis Services** models to support business intelligence and data analytics requirements, creating measures, dimensions, and hierarchies for reporting and visualization.
- Built and configured **Jenkins** slaves for parallel job execution. Installed and configured **Jenkins** for continuous integration and performed continuous deployments. Worked on migration of data from On - prem **SQL** server to Cloud databases (**Azure Synapse Analytics** (DW) & **Azure SOL DB**).
- Designed and deployed a **Kubernetes**-based containerized infrastructure for data processing and analytics, leading to a 20% increase in data processing capacity.
- Skilled in monitoring servers using **Nagios**, **Cloud watch** and using **ELK Stack- Elastic search** and **Kibana**.
- Worked on **Azure Data Factory** to integrate data of both on-prem (**MY SQL**, **Cassandra**) and cloud (**Blob** storage, **Azure SQL DB**) and applied transformations to load back to **Azure Synapse**.
- Built **PySpark** code in validate the data from raw source to **Snowflake tables**.
- Good experience with Continuous Integration and Continuous Delivery (**CI/CD**) of application using **Bamboo**.

Environment: API, Azure, Azure Analysis Services, Azure Synapse Analytics, Blob, Cassandra, CI/CD, Data Factory, Elasticsearch, ETL, Factory, HBase, HDFS, Java, Jenkins, Kafka, Kubernetes, lake, Oracle, Power BI, PySpark, Python, RDBMS, Services, Snowflake, Spark, Spark Streaming, SQL, SSIS, Teradata

Client: American Express, Chennai, India

Sep 2022 - May 2023

Role: AWS Data Engineer

Description: American Express is a leading financial services company specializing in credit cards, charge cards, and travel-related services. Contributed to the development and optimization of data ingestion, ETL processes, and real-time data analysis, enhancing system performance and reliability across various platforms.

Responsibilities:

- Developed custom multi-threaded **Java** based ingestion jobs as well as **Sqoop** jobs for ingesting from **FTP** servers and data warehouses. Worked on developing ETL processes (**Data Stage Open Studio**) to load data from multiple data sources to **HDFS** using **FLUME** and **SQOOP**, and performed structural modifications using **Map Reduce**, **HIVE**.
- Used **Data Build Tool** for transformations in ETL process, **AWS Lambda**, **AWS SQS**
- Created **Hive**, **HBase** tables and **HBase** integrated **Hive** tables as per the design using **ORC** file format and **Snappy** compression. Developed **SQOOP** scripts to migrate data from **Oracle** to **Big data Environment**.
- Wrote **Kafka** producers to stream the data from external rest **APIs** to **Kafka** topics.
- Developed **PySpark/Spark SQL** scripts to analyze various customer **behaviors**.
- Automated and monitored **AWS** infrastructure with **Terraform** for high availability and reliability, reducing infrastructure management time by 90% and improving system uptime.
- Actively Participated in all phases of the Software Development Life Cycle (**SDLC**) from implementation to deployment.
- Wrote and executed various **MYSQL** database queries from **Python** using **Python-MySQL connector** and **MySQL dB** package. Used **Jira** for ticketing and tracking issues and **Jenkins** for continuous integration and continuous deployment.
- Implemented **Azure data lake**, **Azure Data factory** and **Azure** data bricks to move and conform the data from on - premises to cloud to serve the analytical needs of the company.
- Developed **Kibana Dashboards** based on the Log stash data and Integrated different source and target systems into **Elasticsearch** for near real time log analysis of monitoring End to End transactions.
- Performed continuous Integration/Continuous delivery (**CI/CD**) on **Jenkins** build pipeline and fixed failure issues.
- Used **AWS** to create storage resources and define resource attributes, such as disk type or redundancy type, at the service level. Used **Docker** for managing the application environments.

Environment: AWS, CI/CD, Docker, Elasticsearch, ETL, HBase, HDFS, Hive, Java, Jenkins, Jira, Kafka, lake, Lambda, MySQL, Oracle, PySpark, Python, Spark, Spark SQL, SQL, Sqoop

Client: (TCS) Nationwide Insurance, Chennai, India

Apr 2021 - Aug 2022

Role: GCP Data Engineer

Description: Nationwide Mutual Insurance Company is an insurance and financial services companies. Involved in migrating data systems to the cloud, optimizing data processing pipelines, and implementing scalable ETL solutions to enhance data delivery and analytics efficiency.

Responsibilities:

- Got involved in migrating on prem **Hadoop** system to using **GCP** (Google Cloud Platform).
- Wrote scripts in **Hive SQL/Presto SQL**, using python plugin for both **spark** and **presto** for creating complex tables with high performance metrics like partitioning, clustering and skewing. Worked on confluence and **Jira**.
- Migrated previously written **cron** jobs to **airflow/composer** in **GCP**. Leveraged cloud and **GPU** computing technologies for automated machine learning and analytics pipelines, such as **AWS**, **GCP**
- Designed and implemented configurable data delivery pipeline for scheduled updates to customer facing data stores built with **Python**. Measured Efficiency of **Hadoop/Hive** environment ensuring **SLA** is met
- Proficient in **Machine Learning techniques** (Decision Trees, Linear/Logistic Regressors) and **Statistical Modeling**
- Compiled data from various sources to perform complex analysis for actionable results
- Optimized the **Tensorflow** Model for efficiency. Analyzed the system for new enhancements/functionalities and perform Impact analysis of the application for implementing **ETL** changes
- Implemented a **Continuous Delivery pipeline** with **Docker**, and **Git Hub** and **AWS**
- Built performant, scalable **ETL** processes to load, cleanse and validate data
- Have written **python DAGS** in **airflow** which orchestrate end to end data pipelines for multiple applications.
- Was involved in setting up of apache **airflow** service in **GCP**.
- Build data pipelines in **airflow** in **GCP** for **ETL** related jobs using different **airflow** operators.

Environment: GCP, Big Query, Cloud SQL, Cloud Storage, Airflow, BQ-ML, DataStudio, MySQL, Git, Atlassian, Natural Languages, Windows server, python, shell scripts, IAM Security, Service Data Transfer, VPC Configuration, Data Catalog, VPN Google-Client

Client: AstraZeneca, Chennai, India

Jan 2019 - Mar 2021

Role: Data Engineer

Description: AstraZeneca is a global biopharmaceutical company specializing in innovative medicines across oncology, cardiovascular, and respiratory fields. Played a key role in enhancing data processing efficiency and system integration by developing middleware solutions, optimizing data workflows, and managing cluster resources and data pipelines.

Responsibilities:

- Worked on **Java Message Service JMS API** for developing message-oriented middleware **MOM** layer for handling various asynchronous requests. Migrated **Hive** QL queries on structured into **Spark SQL** to improve performance.
- Strong knowledge of **ETL** best practices and experience designing and implementing **ETL** workflows using Talend.
- Managed, Configured and scheduled resources across the cluster using **Azure Kubernetes Service**.
- Worked on Big Data Integration & **Analytics** based on **Hadoop, SOLR, PySpark, Kafka, Storm** and webMethods.
- Have worked on partition of **Kafka** messages and setting up the replication factors in **Kafka Cluster**. Created **Databricks** Job workflows which extracts data from **SQL** server and upload the files to **SFTP** using **PySpark** and python.
- Worked on **Spark** to improve the performance and optimization of the existing algorithms in data bricks using **Spark** context, **Spark.SQL**, **Data Frame**, **pair RDD's**. Processed the schema oriented and non-schema-oriented data using **Scala** and **Spark**. Involved in various phases of Software Development Lifecycle (**SDLC**) of the application, like gathering requirements, design, development, deployment, and analysis of the application.
- Extract **Transform** and Load data from Sources Systems to **Azure Data Storage** services using a combination of **Azure Data Factory**, **T-SQL**, **Spark SQL** and **Azure Data Lake Analytics**. Data Ingestion to one or more **Azure** and processing the data in In **Azure Databricks**.
- Ensured data integrity and consistency during migration, resolving compatibility issues with **T-SQL scripting**.
- Building/Maintaining **Docker** container clusters managed by **Kubernetes Linux**, Bash, **Git**, **Docker**. Implemented a continuous delivery (CI/CD) pipeline with **Docker** for custom application images in the cloud using **Jenkins**.

Environment: Azure, Azure Data Lake, CI/CD, Cluster, Data Factory, Docker, DynamoDB, ETL, Factory, Git, Hive, Java, Jenkins, Kafka, Kubernetes, Lake, Linux, PySpark, S3, Scala, Spark, Spark SQL, SQL, Storm